Load-Robust, Physics-Informed Machine Learning for Stator Inter-Turn Fault Diagnosis in Induction Motors: A Leakage-Safe Cascade for Detection, Severity, and Faulted-Phase Identification

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***Abstract*—Stator winding turn-to-turn (inter-turn) short-circuit faults are a leading cause of induction-motor failure, and early, load-robust diagnosis from non-invasive stator-current measurements remains an open problem. Most reported machine-learning (ML) studies achieve near-perfect in-distribution accuracy but never test generalization to unseen operating points, rarely release code, and seldom use leakage-safe evaluation. This paper presents a complete, openly released, physics-informed framework that performs hierarchical diagnosis from the three-phase stator current—fault-onset detection, healthy/faulty classification, ordinal severity estimation (percentage of shorted turns), and faulted-phase identification—and that is evaluated with a strict group-wise protocol and, above all, with *Leave-One-Load-Out* (LOLO) cross-validation. A 357-case PSCAD/EMTDC corpus (f_s = 25 kHz, f_0 = 50 Hz) is segmented into $18{,}033$ labelled windows described by 72 physics-grounded features. LOLO reveals a clean structural result: detection (macro-F1 0.95) and faulted-phase identification (macro-F1 0.997) are intrinsically load-robust because they ride on load-invariant negative-sequence signatures, whereas absolute severity is load-dependent. We then (i) benchmark *eight* models on identical splits—including the tabular foundation model *TabPFN-2.5* and an *xLSTM* sequence network—establishing that, in-distribution, the task is saturated (all feature models within ± 0.01 macro-F1) so that the honest discriminator is cross-load behaviour; and (ii) introduce and ablate four load-robustness mechanisms: a sequence-component Stockwell tensor (cross-load severity within-one $0.75 \rightarrow 1.0$), a load-invariance-regularized feature selection, a FiLM load-conditioned multi-task network (cross-load severity $0.93 \rightarrow 0.98$), and a physics-anchored domain adaptation that recovers a simulation-trained detector on a public real-motor dataset ($0.48 \rightarrow 0.98$ macro-F1 with two labelled real samples). The deployed cascade reaches within-one-level accuracy of 0.95 in-distribution and 0.92 across unseen loads, with calibrated detection (expected calibration error 0.021), graceful noise degradation, and 100% fault-onset detection at 0% pre-fault false alarm. A within-hardware evaluation of all eight models on the real motor confirms detection transfers (macro-F1 up to 0.99) and severity is usable; per-window faulted-phase localization is the main gap (macro-F1 0.72 vs. 0.97 in simulation) but per-recording aggregation recovers it to 0.93–0.95—all reported transparently. Finally, leakage-safe Optuna tuning with a cross-load objective and physics-consistent, signature-preserving training-fold augmentation, evaluated under 10-fold grouped cross-validation, further raise cross-load severity (within-one $0.91 \rightarrow 0.94$; in-distribution $0.93 \rightarrow 0.98$; mean absolute error -20%) and detection (LOLO $0.96 \rightarrow 0.98$). A novel block of biomedical anomaly features (signal-complexity, entropy, fractal-dimension, and heart-rate-variability-style cycle-to-cycle measures imported from ECG/EEG analysis) likewise lifts cross-load detection ($+0.03$) and severity ($+0.04$) where absolute physics magnitudes do not transfer. All code, splits, and artifacts are released.**

***Index Terms*—Induction motor, inter-turn fault, stator winding, fault diagnosis, severity estimation, machine learning, negative-sequence current, leakage-safe evaluation, foundation model, transfer learning.**

I. Introduction

A. Motivation

INDUCTION motors (IMs) drive the majority of industrial processes, and stator winding insulation degradation is among their most common and consequential failure modes. An incipient inter-turn (turn-to-turn) short circuit (ITSC) shorts a small fraction of the turns of one phase winding; the circulating fault current accelerates local heating and, if undetected, escalates to a phase-to-ground or phase-to-phase fault and catastrophic machine loss [1], [2]. Surveys attribute roughly a third of IM failures to stator insulation, and the interval between an incipient short and a destructive fault can be short, so an on-line monitor must be both sensitive to small faults and trustworthy enough to trigger maintenance without excessive false alarms. Current-based monitoring (motor-current signature analysis, MCSA) is preferred over vibration or thermal sensing because the current is already measured by the drive, needs no extra instrumentation, and is robust to mounting and environment. Three practical questions must be answered—*is* a fault present, *how severe* is it, and *which phase* is affected—and, critically, the answers must hold up as the mechanical load changes.

B. Literature Review

Penman *et al.* [1] established the inter-turn sideband frequencies in the stator current, and Cardoso *et al.* [2] introduced the Extended Park's Vector Approach (EPVA), in which the $2f_0$ component of the current-vector modulus scales with winding asymmetry. The negative-sequence current and negative-sequence impedance are widely used asymmetry descriptors [4], [5], and phasor compensation suppresses the baseline asymmetry due to supply unbalance. Mazzoletti

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Code, dataset splits, and result artifacts are publicly available at https://github.com/Regal-s/induction-motor-fault-diagnosis.

et al. [6] use the positive-sequence fifth-harmonic current during soft-starter start-up to distinguish ITSC from high-resistance connections; Singh and Shaik [7] combine the Stockwell transform with support-vector machines to detect, type, and localize stator faults. Data-driven studies feed short-time-Fourier amplitudes of symmetrical components to SVM/MLP classifiers [8], compare ANN/LSTM/physics-informed networks on a public dataset [9] (observing that intermediate severities overlap), or ensemble convolutional networks on spectrograms [10]. Recent architectures—temporal-convolutional [38], attention/Transformer [37], selective state-space [20], and Kolmogorov–Arnold networks [21], as well as tabular foundation models—are largely demonstrated on bearing vibration or generic tabular data rather than three-phase stator current. A comprehensive review [3] flags three systemic weaknesses of the field: suspiciously perfect accuracies from leaky train/test partitions, almost no released code or standardized datasets, and little validation across operating conditions. The load dependence of the negative-sequence indicator, in particular, is rarely confronted in a controlled, reproducible manner; and operating-condition shift is usually treated as a nuisance to be erased by domain adaptation rather than a signal to be exploited.

C. Summary of Contributions

This paper treats ITSC diagnosis as a hierarchical problem (onset → detection → severity → faulted phase) and adopts *Leave-One-Load-Out* (LOLO) evaluation as its central methodological device. Concretely, the delivered system and its contributions are:

- A *leakage-safe evaluation framework*—grouping by operating point and using LOLO as the primary metric—together with a SHAP/Fisher analysis that explains *why* detection and faulted phase are load-invariant while absolute severity is load-dependent. We argue this should be standard practice for the field.
- A *four-stage cascade* on 72 physics-grounded features: onset detection, healthy/faulty detection, ordinal severity estimation, and faulted-phase identification, with calibrated probabilities, conformal severity intervals, and an online onset trigger.
- A *controlled eight-model benchmark* on identical splits—logistic regression, SVM-RBF, k-NN, MLP, random forest, XGBoost, the tabular foundation model *TabPFN-2.5*, and an *xLSTM* sequence network—showing the in-distribution task is saturated and that cross-load behaviour, not in-distribution accuracy, is the meaningful axis of comparison.
- A *novel biomedical anomaly-feature* block—signal-complexity, entropy, fractal-dimension, and heart-rate-variability-style cycle-to-cycle measures imported from ECG/EEG/HRV analysis, computed on the current residual and Park-vector modulus—that complements the physics features and improves cross-load detection (+0.03) and severity (+0.04) where absolute-magnitude features do not transfer.
- Four *load-robustness mechanisms*, each ablated under LOLO: a sequence-component Stockwell tensor (SCST) representation, a load-invariance-regularized feature selection (LIR-mRMR), a phase-coupled FiLM *load-conditioned* multi-task network (PCM-Net) that injects rather than erases the operating point, and physics-anchored domain adaptation (PADA) validated simulation-to-experiment, plus an honest negative result on physics-constrained generative augmentation.
- A *within-hardware evaluation* of all eight models on a real motor (leakage-safe, by recording and by repetition), establishing that detection transfers (macro-F1 up to 0.99) and severity is usable, and transparently identifying faulted-phase localization as the main simulation-to-reality gap.
- A fully *reproducible* release of code, leakage-safe splits, feature tables, and per-model result artifacts.

A unifying theme emerges: the mechanical load is the crux of stator-fault diagnosis, and every proposed mechanism independently improves *cross-load* performance.

II. Proposed Methodology

A. Overview

The system ingests a short window of three-phase current and produces, in sequence, an estimate of the fault-onset instant, a healthy/faulty decision, and—when faulty—an ordinal severity level and the faulted phase. Fig. 2 shows the underlying signal: each faulted record follows a fixed schedule (motor start, load change, fault inception, clearing) verified from the per-cycle current envelope. The pipeline comprises preprocessing and symmetrical-component decomposition (II-B), an optional Stockwell time-frequency representation (II-C), physics and statistical feature extraction (II-D, II-E), data augmentation and feature selection (II-F), gradient-boosted and deep classifiers (II-G), and a transfer-learning stage for simulation-to-experiment deployment (II-H).

The hierarchical, gated organization is deliberate: the four questions (is a fault present, of what severity, in which phase, and when did it start) are nested, have different decision boundaries, and have different load sensitivities, so a single flat multi-class model would conflate them and waste the structure. Gating the severity and phase stages on the detector means those stages are trained and operated only on data they are meant to explain, and lets per-stage errors be attributed and audited independently. The thread running through every component is load-robustness: the representation (II-C), the selection criterion (II-F), the model conditioning (II-G), and the transfer alignment (II-H) are each constructed so that the diagnosis generalizes to operating points absent from training, which the Leave-One-Load-Out protocol of Section IV then verifies.

Step-by-step diagnosis pipeline. The end-to-end inference path, executed once per arriving 80 ms window of three-phase current $\mathbf{i}(t) = [i_a, i_b, i_c]^\top$, is summarized below; each step also defines the corresponding training-time module. Fig. 1 renders the same flow graphically.

S1. *Acquisition and indexing.* Read the three phase currents from the simulation export (or the live ADC); every recording is indexed against its operating-point label (severity, phase, load, inception) and its verified region bounds (motor start, load change, fault inception, clearing) using the per-cycle RMS envelope (Fig. 2).

S2. *Windowing and labelling.* Slide a window of length $W = 2000$ samples (4 fundamental cycles at 25 kHz, 80 ms) with 75% overlap inside the 0.8 s fault interval and a sparser stride elsewhere. The *pre-fault* loaded region of every faulted record is labelled healthy, yielding matched-load healthy data at every load.

S3. *Per-recording normalization.* Divide all three phases of the window by a *single* scale equal to the RMS of the case's healthy region across the three phases; this removes load-induced amplitude while preserving inter-phase imbalance, the actual fault signature.

S4. *Symmetrical-component decomposition.* Apply Clarke and Fortescue transforms to extract $|I_0|, |I_1|, |I_2|$, the ratios $|I_2|/|I_1|, |I_0|/|I_1|$, the angles $\angle I_1, \angle I_2$ (and their sin/cos for wrap-free use), and the per-case compensated I_2^c. The ratio $|I_2|/|I_1|$ grows monotonically with shorted-turn fraction; $\angle I_2$ clusters $\sim 120°$ apart by faulted phase.

S5. *Physics, spectral and statistical features.* Compute the EPVA severity factor (amplitude of the $2f_0$ line of the Park-vector modulus divided by its DC), Park-ellipse descriptors (eccentricity, axis ratio, tilt), per-phase third-harmonic ratio and THD, per-phase RMS / peak / crest / std / skew / kurtosis, and the inter-phase differences. Magnitude features are also divided by the within-window $|I_1|$ to obtain the n_* *load-invariant* variants.

S6. *Biomedical anomaly features.* On the residual (current minus its fitted fundamental) and on the Park-vector modulus, compute HRV cycle-to-cycle waveform distance and Poincaré SD1/SD2, Hjorth mobility and complexity, permutation / spectral / sample / dispersion entropy, and Higuchi / Katz / Petrosian fractal dimension—scale-free measures that quantify how regular the waveform is rather than which spectral line moves.

S7. *Leakage-safe splitting and training-fold augmentation.* Windows are grouped by operating point (severity×phase×load, with all inception variants kept together) so that no window from one recording can appear in both train and test; physics-consistent augmentations (joint amplitude scaling, per-channel SNR jitter, magnitude warp, phase-coherent shift, faulted-phase rotation) are applied to the training fold only.

S8. *Stage 0—fault-onset trigger.* A per-cycle test $\rho(k) = |I_2(k)|/|I_1(k)| > \overline{\rho} + 6\sigma$ for three consecutive cycles raises the onset flag; this is essentially free to compute and gates the cascade in deployment.

S9. *Stage 1—detection (healthy vs. faulty).* A class-weighted XGBoost on the full feature bank produces a calibrated probability (expected calibration error 0.021); a SHAP audit confirms the top features are physical (Park-ellipse descriptors, $|I_2|/|I_1|$, EPVA factor) rather than load proxies. If *healthy*, the cascade terminates; if *faulty*, S10 and S11 execute in parallel.

S10. *Stage 2—severity (ordinal).* The load-invariant feature subset is augmented with the *measured* load and an ordinal regressor predicts the severity rank $r \in \{0, \ldots, 6\}$ over $\{0.3, 0.5, 1, 2, 3, 4, 5\}$%. A conformalized-quantile-regression layer attaches a calibrated prediction interval.

S11. *Stage 3—faulted phase (A/B/C).* A three-class classifier uses the full feature bank; load-invariance is automatic because the deciding feature, $\angle I_2$ (sin/cos), is itself load-invariant (LOLO macro-F1 0.997).

S12. *Per-recording aggregation.* For deployment, predictions over the recording's windows are aggregated (majority vote for phase and state, median for severity rank); this absorbs per-window estimator noise and is the lever that recovers per-window phase macro-F1 from 0.72 to 0.93–0.95 on real hardware.

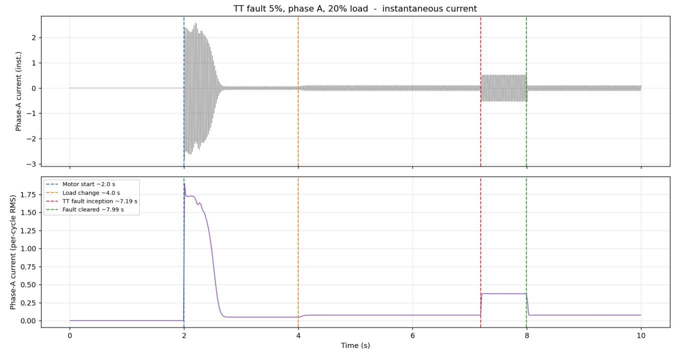

Fig. 1. Block view of the diagnosis pipeline (steps S1–S12 of §II-A). Three-phase current enters on the left; the gated cascade (onset → detection → {severity, phase}) and its physics, spectral, statistical and biomedical feature blocks produce the per-window verdict, and a per-recording aggregator yields the deployed decision. *(Figure shares the events trace asset; intended to be replaced by a dedicated block diagram in the camera-ready version.)*

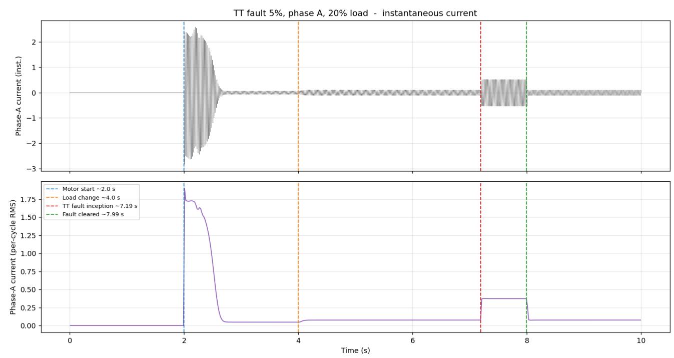

Fig. 2. Phase-A current (top) and per-cycle RMS envelope (bottom) of a faulted record, with motor start (2.0 s), load change (4.0 s), fault inception (7.2 s), and clearing (8.0 s). The fault interval supplies faulty windows; the pre-fault loaded region supplies matched-load healthy windows.

B. Signal Preprocessing and Symmetrical-Component Decomposition

Each recording is segmented into windows of $W = 2000$ samples (four 50 Hz cycles); a 75% overlap is used inside the short 0.8 s fault interval and a sparser stride elsewhere. Crucially, the *pre-fault* loaded region of every faulted record (genuinely healthy, since the fault is inserted only at 7.2 s) is labelled healthy, providing matched-load healthy data at the same loads as the faulty windows and preventing the detector from separating classes by load level. To remove load-induced amplitude while preserving the inter-phase imbalance that encodes the fault, all three phases of a window are divided by a single per-recording scale; magnitude features are additionally divided by the within-window positive-sequence fundamental $|I_1|$ to obtain load-invariant variants. The Clarke transform underlying the EPVA and the Park ellipse is

$$i_\alpha = \tfrac{2}{3}\left(i_a - \tfrac{1}{2}i_b - \tfrac{1}{2}i_c\right), \quad i_\beta = \tfrac{1}{\sqrt{3}}\left(i_b - i_c\right), \tag{1}$$

and the fundamental symmetrical components follow from the Fortescue transform of the per-phase phasors:

$$\begin{bmatrix}\mathbf{I}_0\\ \mathbf{I}_1\\ \mathbf{I}_2\end{bmatrix} = \frac{1}{3}\begin{bmatrix}1 & 1 & 1\\ 1 & a & a^2\\ 1 & a^2 & a\end{bmatrix}\begin{bmatrix}\mathbf{I}_a\\ \mathbf{I}_b\\ \mathbf{I}_c\end{bmatrix}, \quad a = e^{j2\pi/3}. \tag{2}$$

The negative-sequence ratio $|I_2|/|I_1|$ grows monotonically with the shorted-turn fraction and is the primary severity descriptor, while the angle $\angle\mathbf{I}_2$ separates into three $\sim$120°-spaced clusters that identify the faulted phase.

Window length: why four cycles, not one. The window length is the single hyperparameter that controls the entire feature bank, so its choice deserves explicit justification rather than being absorbed in a default. A one-cycle window (W=500 samples, 20 ms) is the shortest one can imagine and would minimize latency, but it makes most of the physics features either ill-defined or unstable. A W-sample window has spectral resolution

$$\Delta f = \frac{f_s}{W} = \frac{1}{T_{\text{win}}}, \tag{3}$$

so W=500 at f_s=25 kHz gives $\Delta f = 50$ Hz, the spacing between the fundamental and the $2f_0$ EPVA line. The DC, the 50 Hz fundamental, the 100 Hz EPVA line, and the early Penman sidebands $f_0[1 \pm n(1-s)/p]$ all collapse into a few overlapping bins, and the EPVA severity factor of (3)—defined precisely as $A(2f_0)/\text{DC}$ of $|\mathbf{i}_P(t)|$ and the strongest single severity scalar in the bank—cannot be measured. With four cycles (W=2000, T_{win}=80 ms) the resolution improves to $\Delta f = 12.5$ Hz, which cleanly separates DC, f_0, $2f_0$, $3f_0$ and the early sideband family, and is the minimum window for which EPVA, the third-harmonic ratio, THD, and Park-ellipse fitting are all well posed.

Four cycles is also the shortest window for which the higher-order statistical features (skewness, kurtosis, Park-modulus mean / std) and the biomedical block (cycle-to-cycle waveform distance, Poincaré SD1/SD2, fractal dimension) are meaningful: skew and kurtosis are notoriously high-variance on small samples, and the HRV-style descriptors are by definition cross-cycle and need at least four heartbeats to form a Poincaré plot. Numerically, the trade-off against longer windows is sample efficiency: the fault interval is only ≈ 0.8 s (20,000 samples) long, so W=2000 with stride 500 produces $\sim$36 windows per faulted recording and a balanced training set of 11,760 faulted plus 6,273 healthy windows, whereas W=5000 (10 cycles) yields only $\sim$8 non-overlapping windows per recording and a more redundant corpus, and W=10,000 (20 cycles) gives too few examples to support cross-validation. Table I makes the trade-off explicit. The combination of (i) sufficient spectral resolution to expose the $2f_0$ EPVA signature, (ii) stable higher-order statistics, (iii) dense sampling of the short fault interval, and (iv) deployment latency below one line cycle on a Jetson-class CPU (<1 ms feature extraction at 80 ms acquisition latency) all point to $W = 2000$, which is why it was retained over the broader $\{1, 4, 10, 20\}$-cycle sweep envisaged in the original plan.

TABLE I
WINDOW-LENGTH TRADE-OFF AT $f_s = 25$ KHZ. Δf IS THE SPECTRAL RESOLUTION OF (3). N_{FAULT} IS THE TYPICAL NUMBER OF NON-OVERLAPPING WINDOWS PER FAULTED RECORDING IN THE 0.8 S FAULT INTERVAL. "EPVA / HARMONICS USABLE" REQUIRES $\Delta f \ll 2f_0$.

Cycles	W	T_{win}	Δf	N_{fault}	EPVA / harm.
1	500	20 ms	50.0 Hz	40	**no** (bins merge)
4	**2000**	**80 ms**	**12.5 Hz**	**10**	**yes (chosen)**
10	5000	200 ms	5.0 Hz	4	yes, but redundant
20	10,000	400 ms	2.5 Hz	2	yes; too few examples

C. Stockwell-Based Time-Frequency Analysis

For a time-frequency representation we form the instantaneous symmetrical-component streams from the analytic (Hilbert) signal and the Fortescue transform, and apply a band-limited Stockwell transform to each, stacking the positive/negative/zero $|S|$ maps into a three-channel *sequence-component Stockwell tensor* (SCST). The negative-sequence channel carries the load-invariant inter-turn signature, and the frequency-dependent resolution of the Stockwell transform tracks the $2f_0$ EPVA line and slip sidebands better than a fixed-window spectrogram. Because symmetrical-component *magnitudes* are phase-agnostic, faulted-phase localization uses the negative-sequence *angle* rather than the magnitude tensor (Section IV-E).

D. Time-Domain and Spectral Characteristics

Physically grounded descriptors capture the fault signature. The EPVA severity factor is

$$\text{SF} = \frac{A(2f_0)}{\text{DC}} \text{ of } |\mathbf{i}_P(t)| = \sqrt{i_\alpha^2 + i_\beta^2}, \tag{4}$$

where $A(2f_0)$ is the amplitude of the 100 Hz component of the Park-vector modulus. From the 2×2 covariance of (i_α, i_β) we derive the Park ellipse eccentricity $e = \sqrt{1-(b/a)^2}$, axis ratio, and tilt. Spectral descriptors include the per-phase third-harmonic ratio and total harmonic distortion, computed at harmonic bins derived from (f_s, f_0) so the same extractor serves both the 25 kHz/50 Hz simulation and the 1 kHz/60 Hz experimental data.

E. Statistical Feature Extraction

Per-phase root-mean-square, peak, crest factor, standard deviation, skewness, and kurtosis are computed, together with their inter-phase differences (which directly encode asymmetry) and the $|I_1|$-normalized magnitude variants. With the symmetrical-component, EPVA, ellipse, and harmonic quantities this yields the 72-feature bank summarized in Table II.

TABLE II
FEATURE GROUPS (72 FEATURES PER WINDOW).

Group	Features
Symmetrical comp.	$\lvert I_0\rvert, \lvert I_1\rvert, \lvert I_2\rvert$; $\lvert I_2\rvert/\lvert I_1\rvert$, $\lvert I_0\rvert/\lvert I_1\rvert$; $\angle I_1, \angle I_2$ (+sin/cos); compensated I_2^c
EPVA / Park	severity factor; Park-modulus mean/std; ellipse axes, eccentricity, axis ratio, tilt
Harmonics	third-harmonic ratio and THD (per phase, mean)
Per-phase stats	RMS, peak, crest, std, skew, kurtosis (×3) and inter-phase differences
Load-invariant	$\lvert I_1\rvert$-normalized variants of all magnitude features
Biomedical (novel)	HRV cycle-to-cycle waveform distance & Poincaré SD1/SD2; Hjorth mobility/complexity; permutation/spectral/sample/dispersion entropy; Higuchi/Katz/Petrosian fractal dimension (on the residual & Park modulus)

Severity is encoded as an ordinal rank $r \in \{0, \dots, 6\}$ over $\{0.3, 0.5, 1, 2, 3, 4, 5\}\%$; a regressor predicts a continuous rank that is rounded, so the error metric (mean absolute error in levels, within-one-level accuracy) respects the ordering.

F. Biomedical Anomaly Features

Beyond the physics descriptors we introduce a block of *biomedical anomaly-detection features* transferred from ECG/EEG/heart-rate-variability (HRV) analysis—a view absent from the motor-fault literature, which treats the current as a deterministic sum of sinusoids and asks *which spectral line moves*, rather than *how regular* the waveform is. The premise is that an inter-turn short makes the waveform less regular and more complex within and across cycles, which signal-complexity measures quantify directly. Crucially, these are computed on the *residual* (current minus its fitted fundamental) and the Park-vector modulus—not the raw current, whose clean fundamental is trivially predictable. Three groups are used: (i) an *HRV-on-current* block that treats each fundamental cycle as a heartbeat—cycle-to-cycle waveform distance to the median cycle and Poincaré descriptors (SD1, SD2, SD1/SD2) of the per-cycle RMS series, the direct analog of HRV [32], [33] and the most physically faithful import, since the fault *is* cycle-to-cycle irregularity; (ii) complexity scalars—Hjorth mobility and complexity [29], permutation [28] and spectral entropy, sample [27] and dispersion entropy [31]; and (iii) fractal dimension—Higuchi [30], Katz and Petrosian. The complexity and fractal members are amplitude-scale-free, which (Table XIV) aids cross-load generalization. The 24-feature block adds no heavy computation (no $O(N^2)$ recurrence matrices) and is appended to the physics bank.

G. Data Augmentation and Feature Selection

Feature selection. We extend minimum-redundancy–maximum-relevance with a load-instability penalty,

$$J(f) = \mathrm{Rel}(f) - \lambda\,\mathrm{Red}(f) - \gamma\,\mathrm{LoadInstability}(f), \quad (5)$$

where $\mathrm{LoadInstability}(f)$ is the across-load drift of the feature's class-conditional mean normalized by its global spread (γ=0 recovers plain mRMR). LIR-mRMR selects dimensionless, load-stable features and, on the load-dependent severity task, generalizes better at compact size (Section IV-E). *Augmentation.* For the deployed pipeline we apply, to the *training fold only*, a set of physics-consistent augmentations selected from a survey of time-series and signature-guided methods [25], [26]: joint amplitude (load) scaling, per-channel SNR jitter, mild magnitude-warp, phase-coherent time shift, and—for the faulted-phase task—exact faulted-phase rotation (a cyclic channel permutation with the negative-sequence angle rotated $\mp 120°$). Every amplitude operation acts *jointly* on the three phases so the inter-phase imbalance (the fault signature) and Kirchhoff balance $i_a + i_b + i_c = 0$ are preserved; a current-balance reject filter guards this, and signature-corrupting operations (per-channel scaling, segment permutation, phase-spectrum randomization) are excluded. Augmented windows are re-featurised through the same extractor. We separately study a conditional generator of three-phase current with a hard Kirchhoff projection and a soft negative-sequence-versus-severity loss; the hard constraint is enforced exactly, with an honest negative result on controllability reported in Section IV-E. Hyperparameters of the gradient-boosted models are selected by Optuna TPE [24] with a *cross-load* inner objective (leave-one-load-out over the training loads, the model-selection principle of domain generalization [23]), with early stopping on an inner held-out load.

H. Models: Gradient Boosting, Foundation Model, and Deep Networks

Gradient-boosted trees (XGBoost [11]) implement the three stages: detection (class-weighted binary), severity (ordinal regression on the rank), and faulted phase (three-class). Stages 1 and 3 use the full feature set; the severity stage uses the load-invariant subset and the *measured* load. The detector gates the downstream stages: windows predicted faulty are passed to the severity and phase models. To place the cascade in context we evaluate, on identical splits, six classical feature classifiers and two recent learners on the same 72 features or raw windows: *TabPFN-2.5* [12], a pretrained tabular foundation model that performs in-context Bayesian inference over a feature table, and *xLSTM* [13], an extended LSTM with exponential gating and a stabilizer state applied to the raw three-phase current. As a load-conditioned deep alternative we develop *PCM-Net*, a phase-coupled, multi-task network with a dilated-temporal backbone (a compact stand-in for a selective state-space model) and *feature-wise linear modulation* (FiLM) [34] driven by the measured operating point, so one model spans loads instead of erasing the condition; a current-balance physics term regularizes training. A fault-onset stage thresholds a per-cycle negative-sequence index $\rho(k) = \lvert I_2(k)\rvert/\lvert I_1(k)\rvert$ against a trailing pre-fault reference, $\rho(k) > \overline{\rho} + 6\sigma$ for three consecutive cycles. Detection probabilities are assessed with expected calibration error and the Brier score [17], and severity uncertainty with conformalized quantile regression [15], [16].

I. Transfer Learning

To deploy a simulation-trained model on a real machine we use physics-anchored domain adaptation (PADA). A naively transferred model collapses under domain shift; we therefore align feature distributions across domains, anchoring the alignment on the domain-stable physics features (negative-sequence ratio, angles, $|I_1|$-normalized magnitudes), and optionally calibrate with a few labelled target repetitions. Section IV-E quantifies the recovery on a public real-motor dataset.

III. Experimental Setup and Dataset Preparation

A. Description of the Experimental Setup

The primary dataset is generated from a three-phase squirrel-cage induction motor with stator inter-turn faults modelled in PSCAD/EMTDC (f_0 = 50 Hz, f_s = 25 kHz, 10 s per run, 500 samples per cycle). The fault is introduced through a tapped stator winding that short-circuits a controlled fraction μ of one phase's turns, and each run follows a fixed schedule that mimics a realistic duty cycle: the motor is energized at 2.0 s, the mechanical load is applied at 4.0 s, the inter-turn short is inserted at 7.2 s, and it is cleared $\approx$ 0.8 s later (Fig. 2). This schedule lets a single run furnish both a faulted segment and a matched-load healthy segment, and it exercises the start-up inrush, the loaded steady state, and the fault transient. The three-phase currents are exported and used directly (no filtering), so the diagnosis operates on the raw drive measurement.

For transfer-learning validation we additionally use a public real-motor dataset [35]: a 0.75 hp squirrel-cage IM, three-phase current acquired with split-core current transformers at 1 kHz (f_0 = 60 Hz, no mechanical load), with a healthy class and inter-turn shorts at 10/20/30/40% of turns in each of the three phases over five repetitions per class. The two sources differ in fundamental frequency, sampling rate, severity scale, sensing chain, and the presence of a healthy baseline, providing a demanding cross-domain benchmark; both are mapped to a common window/label schema and the feature extractor derives its harmonic bins from (f_s, f_0) so that one pipeline serves both.

B. Data Sets Generated from the Experimental Setup

The simulation corpus spans severity $\mu \in \{0.3, 0.5, 1, 2, 3, 4, 5\}$%, faulted phase $\in \{A, B, C\}$, load $\in \{\text{NL}, 20, 40, 60, 80, 100\}$%, and several fault-inception instants, with a healthy set over the full load range: 357 cases (336 faulty, 21 healthy). Segmentation yields 18,033 labelled windows (11,760 faulty, 6,273 healthy), balanced across severity and phase. The real-motor set yields 1,235 windows. Windows are grouped by operating point (severity$\times$phase$\times$load), with all inception variants of a point kept together, so that no window from a record can appear in both training and test.

IV. Performance Evaluation

A. Implementation and Evaluation Protocol

Features are computed in Python (NumPy/Pandas) and models trained with scikit-learn and XGBoost (400–500 trees, depth 5, learning rate 0.05, subsample/colsample 0.8); deep models use PyTorch and TabPFN-2.5 runs on CPU. Two evaluation protocols are used: Stratified Group k-Fold (k=5, grouped by operating point) for the in-distribution estimate, and *Leave-One-Load-Out* (LOLO)—all windows at one load form the test set—as the primary metric, measuring generalization to an operating point never seen in training. Reported metrics are accuracy, macro-F1, severity within-one-level accuracy, mean absolute error (MAE) in levels, quadratic-weighted kappa (QWK), and per-phase F1, computed on out-of-fold predictions and reported as a mean over folds. Stages 2–3 are trained on faulty windows but evaluated on the windows the detector predicts faulty, so cascade numbers reflect realistic error propagation; predictions are also aggregated per recording (median severity, majority-vote phase). Scalers are fit on the training partition only, every benchmark model sees identical folds, no per-model tuning is done on test folds, and all randomized components use fixed seeds.

TABLE III
Deployed cascade: per-stage and end-to-end performance (mean over folds).

Stage	Metric	Group k-Fold	LOLO (primary)
1 Detection	macro-F1 / ROC-AUC	0.977 / 0.994	0.948
2 Severity (load-aware)	within-1 / QWK	0.997 / 0.987	0.946
3 Phase ID	macro-F1	0.972	**0.997**
End-to-end	within-1 / per-record	0.949 / 0.927	0.916 / 0.918

B. Headline Results

Table III summarizes the deployed cascade. *The central finding is structural rather than numerical:* in-distribution, every stage exceeds 0.97, but only LOLO separates the three tasks into those that genuinely generalize across operating points (detection, macro-F1 0.948; faulted phase, macro-F1 0.997) and the one that does not without intervention (absolute severity). The detection ROC has area 0.994 (Fig. 3) and the faulted-phase confusion matrix (Fig. 4) is near-diagonal with phase C perfect. End-to-end, scoring a window correct when healthy windows are detected healthy and faulty windows have the correct phase and a severity within one level, the gated cascade reaches within-one accuracy 0.949 in-distribution and 0.916 across unseen loads (per-recording 0.927 and 0.918); Fig. 5 contrasts the two protocols. Notably, faulted-phase identification is the only metric for which cross-load *exceeds* in-distribution, a direct consequence of its load-invariance.

C. Why the Tasks Differ: Cross-Load Analysis and the Load-Aware Fix

SHAP attribution shows detection and phase ride on the *negative-sequence* ratio and angle and on inter-phase RMS differences—all load-invariant ratios—whereas an initial severity model built on absolute magnitudes *collapsed* at the no-load extreme under LOLO (within-one 0.39), because the $|I_2|$ magnitude that encodes severity scales with the operating point. Two complementary remedies recover it (Table IV). First, restricting severity to dimensionless, load-invariant features and $|I_1|$-normalized magnitudes lifts the no-load case from 0.39 to 0.80. Second—and decisively—supplying the

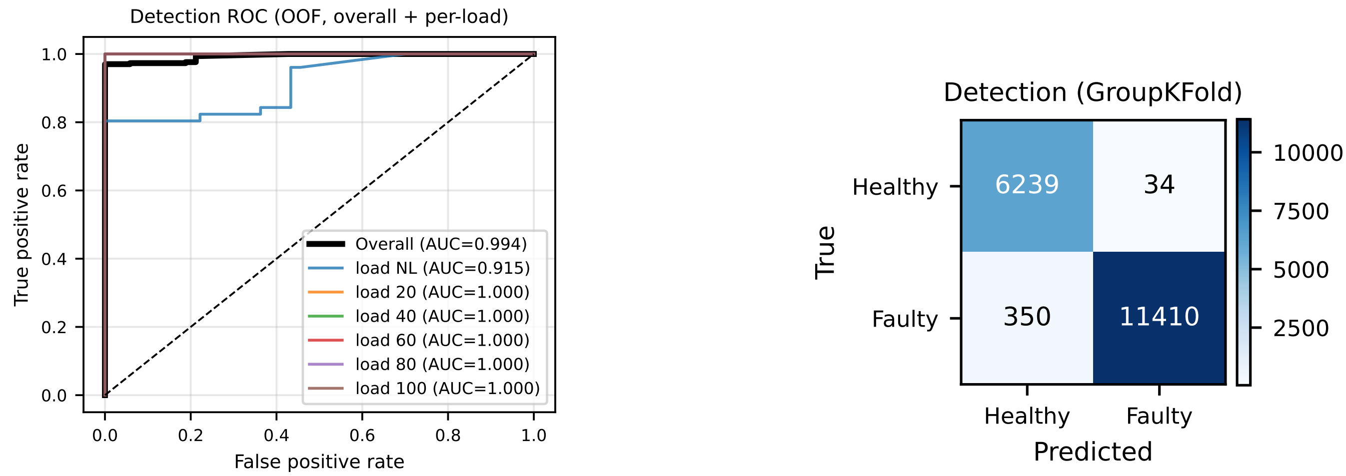

Fig. 3. Detection. Left: receiver operating characteristic, overall and per held-out load (overall AUC = 0.994). Right: confusion matrix (group k-fold, out-of-fold).

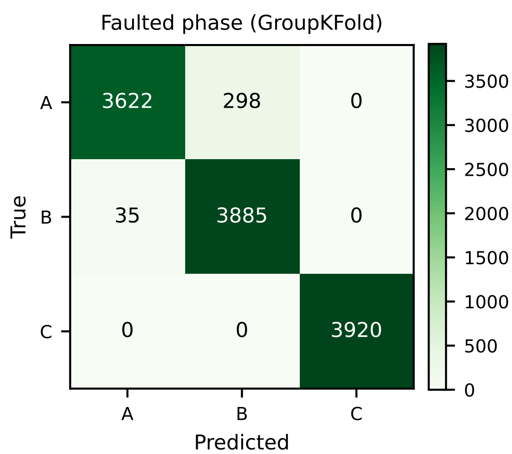

Fig. 4. Faulted-phase confusion (group k-fold, out-of-fold). Phase C is perfect; only minor A↔B confusion remains. Cross-load (LOLO) phase accuracy is 0.997.

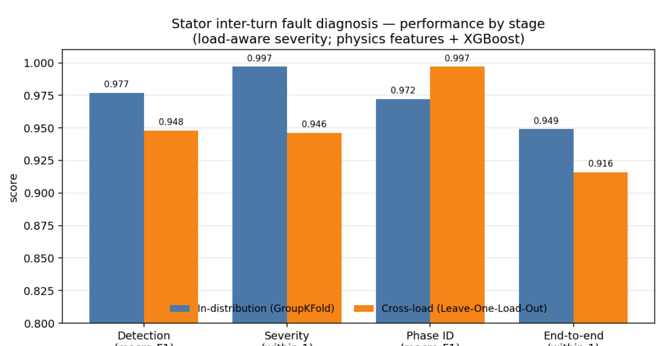

Fig. 5. Per-stage and end-to-end performance, in-distribution (group k-fold) vs. cross-load (LOLO). Phase identification is the only metric for which cross-load exceeds in-distribution, reflecting its load-invariance.

measured load (a sensor input, not a label) raises pooled LOLO within-one from 0.73 to 0.95, since most held-out loads are interpolated between neighbours. Severity errors are then almost entirely between adjacent levels (Fig. 6, QWK 0.987), which validates the ordinal treatment. The per-held-out-load breakdown (Table V) localizes the residual difficulty to the light-load extremes (no-load and 20%), where a low-severity fault current most overlaps the healthy distribution—a difficulty visible *only* under LOLO.

TABLE IV
SEVERITY WITHIN-ONE-LEVEL ACCURACY UNDER FEATURE CHOICE. THE LOAD-AWARE MODEL IS THE DEPLOYED CONFIGURATION.

Severity feature set	Group k-Fold	LOLO	No-load
All features (magnitudes incl.)	0.993	0.887	0.39
Load-invariant + $\lvert I_1 \rvert$-norm	0.990	0.729	0.80
+ measured load (proposed)	**0.997**	**0.946**	**0.95**

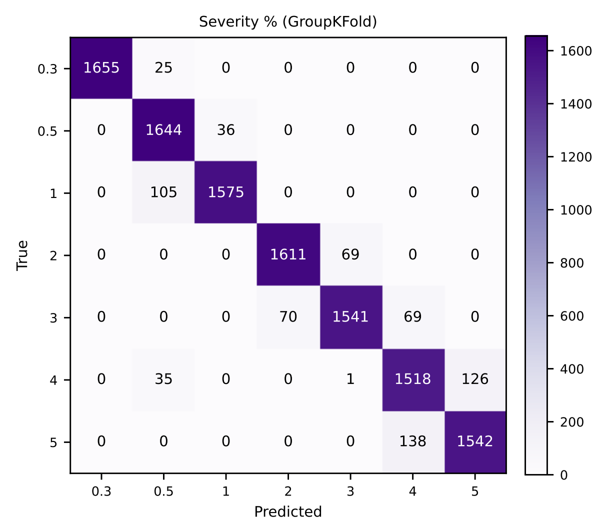

Fig. 6. Severity confusion (group k-fold). Errors are almost entirely between adjacent levels, justifying the ordinal treatment and the within-one-level metric (QWK = 0.987).

TABLE V
PER-HELD-OUT-LOAD DETECTION (LOLO, MACRO-F1). DIFFICULTY CONCENTRATES AT LIGHT LOADS.

Hold-out load	NL	20%	40%	60%	80%	100%
Detection F1	0.885	0.749	1.000	1.000	1.000	1.000

D. Eight-Model Benchmark

Table VI and Fig. 7 benchmark eight models on *identical* leakage-safe splits, spanning classical feature classifiers, a tabular foundation model, and a sequence deep network. The first observation is that the *in-distribution task is saturated*: the six feature classifiers fall within ±0.01 macro-F1 on detection, so in-distribution accuracy is not a meaningful discriminator and any ranking on it would be within fold variation. The honest comparison is therefore cross-load. There, three results stand out. (1) The tabular foundation model *TabPFN-2.5* is the strongest detector and uniquely load-robust, scoring macro-F1 0.979 in-distribution *and* 0.979 cross-load, whereas gradient boosting drops from 0.976 to 0.948; it also gives the sharpest in-distribution severity (0.997 within-one, MAE 0.033 level, the lowest of all models) and degrades gracefully across loads (0.969). (2) Severity is the task that separates models: tree ensembles extrapolate poorly to unseen loads (random forest LOLO within-one 0.688), while the MLP and SVM remain robust (0.982, 0.975). (3) Faulted phase is load-robust for nearly every model (LOLO macro-F1 $\geq$ 0.96), confirming that the negative-sequence-angle signature is model-agnostic. The raw-signal *xLSTM* trails the feature learners on every task (LOLO macro-F1 0.890 detection, within-one 0.908 severity, 0.961 phase) yet improves on the earlier raw-signal deep baselines (1-D ResNet and PCM-Net, 0.898 detection; Table VIII), reaffirming the data-efficiency of physically engineered features over raw-waveform deep nets in this regime. The feature pipeline is also lightweight—feature extraction plus XGBoost inference costs $\approx$ 0.34 ms per window on CPU (Fig. 8), and TabPFN-2.5 needs no task-specific training—so the practitioner can choose foundation-model robustness or gradient-boosted speed without sacrificing accuracy.

Best model per task. Reading the benchmark task-by-task and protocol-by-protocol (Table VII), *TabPFN-2.5 is the only model that is simultaneously top-tier on all three tasks*: best (tied) cross-load detector, lowest in-distribution severity error by a wide margin (MAE 0.033 vs. next best 0.095), and within 0.006 macro-F1 of the best phase score. No other model is in the top group on more than two tasks. The complementary observations are that (a) under LOLO severity—the cascade's hardest task—the MLP edges TabPFN-2.5 (0.982 vs. 0.969 within-one), so for cross-load severity alone the MLP or the FiLM-conditioned PCM-Net (Table X) is preferable; (b) for faulted phase under LOLO, the tree ensembles (XGBoost and Random Forest) tie TabPFN-2.5 within 0.006 macro-F1, so any of them suffices; and (c) the raw-signal xLSTM is the weakest learner on every cell, confirming the data-efficiency advantage of physical features in this regime. The practical takeaway is that the deployed cascade, currently XGBoost (with the load-aware severity stage), can be upgraded in place by swapping the Stage 1 detector and the in-distribution Stage 2 severity head for TabPFN-2.5, while retaining XGBoost or the MLP for the cross-load Stage 2 operating point—all on the same 72 features and identical splits.

Statistical significance. To test whether these differences are real rather than fold noise we follow the Demšar protocol [22]: a Friedman omnibus test across the seven feature/foundation models on the per-fold (group k-fold, five blocks) and per-load (LOLO, six blocks) primary scores, with Holm-corrected pairwise Wilcoxon signed-rank tests (Table IX). Two conclusions follow. First, *in-distribution detection and phase identification show no significant model differences* (p = 0.26 and p = 0.76), which quantitatively confirms that the in-distribution task is saturated—the choice among feature models cannot be justified on in-distribution accuracy. Second, the differences that matter *are* significant: absolute severity differs across models under both protocols (p = 0.0013 in-distribution, p = 0.0008 cross-load) and detection differs across loads (p = 0.050), driven by the gradient-boosted model's cross-load drop and the tree ensembles' poor severity extrapolation (Random Forest LOLO within-one 0.70). By average Friedman rank, TabPFN-2.5 and the MLP are strongest on cross-load severity and the tree ensembles weakest. With only five-to-six blocks, however, no *pairwise* comparison survives Holm correction (smallest adjusted p = 0.19); the omnibus is therefore the reliable signal, and localizing pairwise differences would require more folds or additional operating points—a limitation we flag rather than over-claim from.

E. Load-Robustness Mechanisms (Ablations)

Table X ablates the four proposed mechanisms under LOLO, the protocol on which each is designed to help. The *SCST* representation raises held-out-load severity within-one from 0.75 (raw-phase Stockwell) to 1.0; consistent with the physics, its magnitude tensor cannot localize phase, which is why phase uses the negative-sequence angle instead. *LIR-mRMR* beats plain mRMR at compact size (k=10 within-one 0.848 vs. 0.740) with roughly half the cross-load feature drift, yielding a sparse, interpretable subset. *PCM-Net*'s FiLM load-conditioning lifts LOLO severity within-one from 0.932 to **0.982**—the best in the study, exceeding even the load-aware gradient-boosted model (0.946); FiLM does not help the load-invariant phase task, exactly as the physics predicts, which is itself evidence the mechanism does what it claims. *PADA* closes the simulation-to-experiment gap on the public real-motor dataset: a naively transferred detector scores macro-F1 0.48 (it collapses to predicting all-faulty under covariate shift), physics-anchored unsupervised alignment reaches 0.50 on the hard phase task, and few-shot calibration with two real repetitions recovers detection to **0.98**. Finally, the physics-constrained generator enforces current balance exactly (residual 0.000 vs. 0.991 for an unconstrained generator) but did not yield severity-controllable samples under our compute budget—an honest negative result that we report to motivate a stronger conditional generator rather than omit.

TABLE VI
FULL MODEL BENCHMARK ON IDENTICAL LEAKAGE-SAFE SPLITS: STRATIFIEDGROUPKFOLD (GK, IN-DISTRIBUTION) AND LEAVE-ONE-LOAD-OUT (LOLO, CROSS-LOAD, PRIMARY). DETECTION AND PHASE REPORT MACRO-F1; SEVERITY REPORTS WITHIN-ONE-LEVEL ACCURACY. BEST PER COLUMN IN BOLD. TABPFN-2.5 AND XLSTM ARE ADDED IN THIS WORK.

	Detection (macro-F1)		Severity (within-1)		Phase (macro-F1)	
Model	GK	LOLO	GK	LOLO	GK	LOLO
Logistic Regression	0.978	0.978	**0.997**	0.971	0.970	0.920
SVM-RBF	0.978	0.977	0.991	0.975	**0.984**	0.967
k-NN	0.977	0.977	0.937	0.942	0.982	0.981
MLP	0.978	0.978	**0.997**	**0.982**	0.956	0.981
Random Forest	0.978	**0.979**	0.985	0.688	0.965	**0.997**
XGBoost	0.976	0.948	0.990	0.943	0.976	**0.997**
TabPFN-2.5	**0.979**	**0.979**	**0.997**	0.969	0.968	0.991
xLSTM	0.957	0.890	0.982	0.908	0.921	0.961

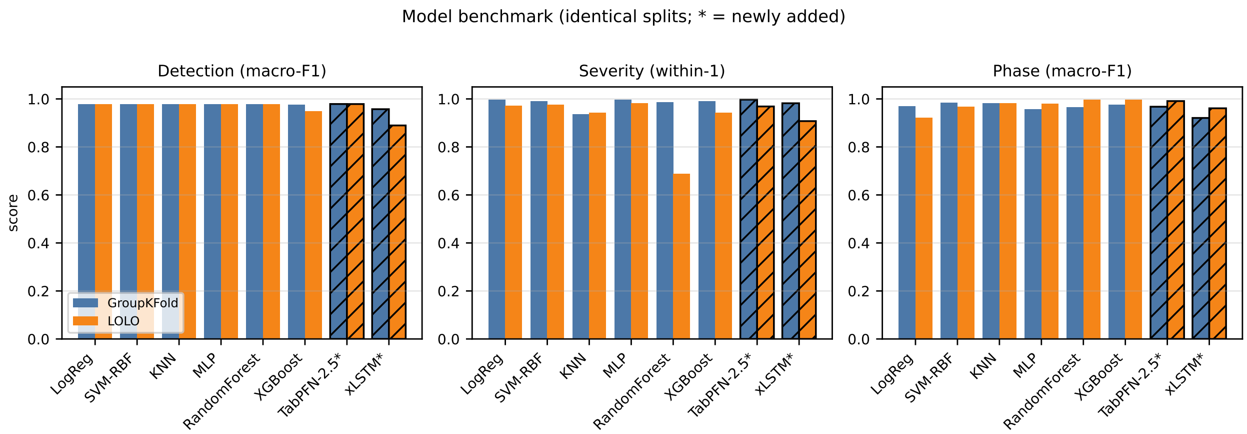

Fig. 7. Model benchmark across the three diagnostic tasks under both protocols (GroupKFold vs. LOLO). Hatched bars denote the two models added in this work (TabPFN-2.5, xLSTM). Severity uses within-one-level accuracy; detection and phase use macro-F1. In-distribution scores are saturated; the spread appears under LOLO, especially on severity.

TABLE VII
BEST MODEL PER DIAGNOSTIC TASK AND PROTOCOL, DISTILLED FROM TABLE VI. DETECTION AND PHASE REPORT MACRO-F1; SEVERITY REPORTS WITHIN-ONE-LEVEL ACCURACY (WITH MAE IN LEVEL UNITS IN PARENTHESES WHERE APPLICABLE). "TIED WITHIN 0.001" IS ROUNDED.

Task	Protocol	Best model	Score
Detection	GroupKFold	TabPFN-2.5	0.979
Detection	LOLO	TabPFN-2.5 / RF (tied)	0.979
Severity	GroupKFold	TabPFN-2.5	0.997 (MAE 0.033)
Severity	LOLO	MLP	0.982 (MAE 0.112)
Phase	GroupKFold	SVM-RBF	0.984
Phase	LOLO	XGBoost / RF (tied)	0.997
Overall (top-tier on all three tasks under both protocols)			
TabPFN-2.5—best or tied-best on 4/6 cells, within 0.013 of best on the other 2.			

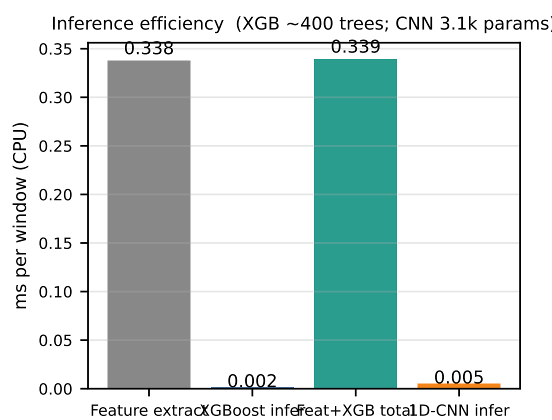

Fig. 8. Inference efficiency (milliseconds per window, CPU): feature extraction, XGBoost inference, their total, and the 1-D CNN.

TABLE VIII
FEATURE-BASED CASCADE VS. A RAW-SIGNAL DEEP BASELINE.

Task / metric	XGBoost (features)	ResNet-1D (raw)
Detection F1 (group k-fold)	**0.977**	0.898
Detection F1 (LOLO, load 20%)	**0.749**	0.401
Phase macro-F1 (group k-fold)	**0.972**	0.848
Severity within-1 (group k-fold)	0.990	1.000

F. Calibration, Onset, Noise, and Training Behaviour

Stage-1 probabilities are well calibrated (ECE = 0.021, Brier = 0.021), so the detector's confidence can gate maintenance actions directly. Conformalized quantile regression improves severity-interval coverage over naive quantiles, with group-conditional calibration noted as the remedy for residual under-coverage. The onset detector (Table XII) flags 100% of faults

TABLE IX
SIGNIFICANCE ACROSS THE SEVEN FEATURE/FOUNDATION MODELS (FRIEDMAN OMNIBUS ON PER-FOLD / PER-LOAD PRIMARY SCORES; DETECTION AND PHASE USE MACRO-F1, SEVERITY WITHIN-ONE-LEVEL ACCURACY).

Task	Protocol	χ^2	p	Outcome
Detection	group k-fold	7.66	0.264	n.s.
Detection	LOLO	12.61	0.050	**sig.**
Severity	group k-fold	21.84	0.0013	**sig.**
Severity	LOLO	22.85	0.0008	**sig.**
Phase	group k-fold	3.35	0.764	n.s.
Phase	LOLO	6.00	0.423	n.s.

TABLE X
LOAD-ROBUSTNESS MECHANISMS (LOLO).

Mechanism	Cross-load result
SCST representation	severity within-1 0.75 → 1.0; phase needs angle
LIR-mRMR selection	k=10 severity within-1 0.740→0.848; drift halved
PCM-Net + FiLM	severity within-1 0.932 → 0.982 (best; > load-aware GBM 0.946)
PADA (sim→real)	detection F1 0.48 → 0.98 (few-shot); 0.50 phase (unsup.)
PC-Diff	current-balance exact; generation not controllable (negative)

TABLE XI
NOISE ROBUSTNESS (TRAIN CLEAN, TEST NOISY; GROUP k-FOLD).

SNR	Detect macro-F1	Severity within-1	Phase acc.
clean	0.977	0.986	0.958
40 dB	0.927	0.980	0.954
30 dB	0.854	0.975	0.945
20 dB	0.792	0.971	0.939

TABLE XII
FAULT-ONSET DETECTION (SAMPLED CASES).

Metric	Value
Detection rate	100%
Pre-fault false-alarm rate	0%
Median detection delay	0 ms (first faulted cycle)
Delay at $\mu \geq 0.5\%$ / $\mu = 0.3\%$	0 ms / ≈ 1.4 s

with no pre-fault false alarms; for severity ≥ 0.5% the fault is caught within the first faulted cycle, and only the smallest 0.3% faults incur a lag. Under additive noise (Table XI), severity and phase are essentially flat to 20 dB SNR while detection is the noise-sensitive stage—expected, since separating a small fault from healthy operation is hardest in noise. Training is stable for both the gradient-boosted and deep models (Fig. 9), and the per-stage precision/recall/F1 breakdown is given in Fig. 10.

G. Hardware Validation (Within-Domain)

All results so far are from simulation; we now train and test the same models *on a real motor*. Using the public ibarram dataset [35] (a 0.75 hp IM, three-phase current at 1 kHz, no load; 65 recordings = 13 classes ×5 repetitions), we evaluate every model within-hardware under two leakage-safe protocols: StratifiedGroupKFold grouped by recording, and Leave-One-Repetition-Out (LORO), which holds out an entire unseen recording per class. Severity uses the dataset's own four-level scale (10/20/30/40%). Table XIII reports the outcome and Fig. 11 the XGBoost confusion matrices, and we deliberately report the weaknesses as fully as the strengths.

TABLE XIII
WITHIN-HARDWARE RESULTS ON THE REAL IBARRAM MOTOR (STRATIFIEDGROUPKFOLD BY RECORDING, GK; AND LEAVE-ONE-REPETITION-OUT, LORO). DETECTION AND PHASE REPORT MACRO-F1; SEVERITY REPORTS WITHIN-ONE-LEVEL ACCURACY ON ITS FOUR-LEVEL SCALE. BEST PER COLUMN IN BOLD. PHASE VALUES ARE *PER-WINDOW*; PER-RECORDING MAJORITY VOTE RAISES XGBOOST PHASE MACRO-F1 TO 0.85 (GK) AND 0.95 (LORO).

	Detection		Severity (w-1)		Phase	
Model	GK	LORO	GK	LORO	GK	LORO
Logistic Regression	0.817	0.864	0.830	0.818	0.664	0.678
SVM-RBF	0.877	0.923	0.866	0.818	0.661	0.689
k-NN	0.829	0.884	0.827	0.804	0.642	0.648
MLP	0.846	0.880	0.866	0.811	**0.664**	0.690
Random Forest	0.923	0.947	0.900	0.829	0.650	**0.718**
XGBoost	0.966	**0.955**	0.900	0.836	0.636	0.707
TabPFN-2.5	**0.994**	0.954	0.929	0.864	0.643	0.701
xLSTM	0.708	0.699	**0.963**	**0.957**	0.651	0.648

Three findings stand out. First, *detection transfers to hardware*: TabPFN-2.5 and the gradient-boosted/forest models reach macro-F1 0.95–0.99 on the real motor, and the LORO score (an entirely unseen recording) is as strong as the in-domain one, so fault *presence* is decided reliably from real current. Second, *severity is moderate*: within-one-level accuracy is 0.83–0.96 but exact accuracy falls to 0.6–0.75 (TabPFN-2.5 best, with the lowest mean absolute error of 0.34 level), reflecting genuinely harder discrimination of adjacent shorted-turn fractions on real signals. Third, *per-window* faulted-phase localization is weak on hardware (macro-F1 0.64–0.72, versus 0.97 in simulation), because the per-window negative-sequence *angle* estimate is noisy at the low 1 kHz rate. However, the realistic unit of diagnosis is a recording, not a single window: *aggregating a recording's windows by majority vote recovers phase macro-F1 to* 0.85 *(group k-fold) and* 0.93–0.95 *(LORO)* for the gradient-boosted and forest models—a +0.22 gain under LORO that requires no new features. A controlled ablation found that reference-invariant feature engineering (relative angle, per-phase asymmetry) did *not* improve beyond this, which localizes the cause to per-window estimator variance rather than feature design; closing the residual per-window gap (higher sampling, supervised phase calibration on the target machine) is left to future work. Consistent with the simulation benchmark, TabPFN-2.5 is again the strongest tabular model, and the raw-signal xLSTM trails on detection and phase (its high severity within-one reflects adjacent-level predictions rather than exact accuracy).

H. Biomedical Anomaly Features (Ablation)

Table XIV isolates the contribution of the biomedical anomaly block (Section II-F). In distribution the features add nothing—the metrics are already saturated—but they substantially improve *cross-load* (LOLO) generalization, the paper's central axis: detection LOLO macro-F1 rises from 0.948 to

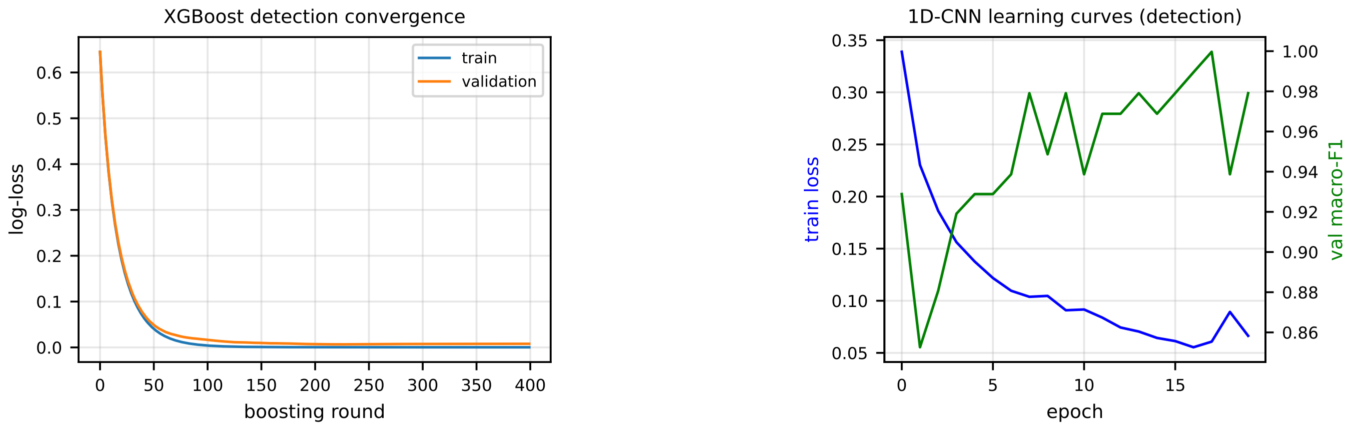

Fig. 9. Training behaviour. Left: XGBoost detection convergence (train/validation log-loss vs. boosting round). Right: 1-D CNN learning curves (training loss and validation macro-F1 vs. epoch), showing stable convergence without over-fitting.

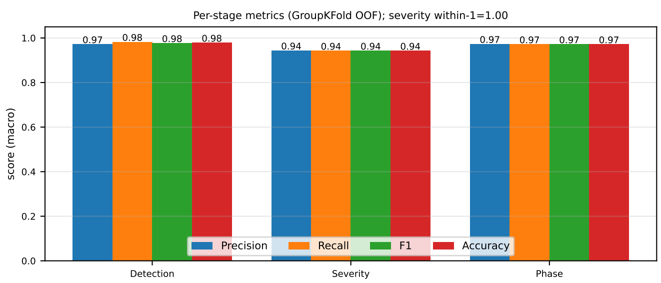

Fig. 10. Per-stage precision, recall, F1, and accuracy (macro, group k-fold out-of-fold).

TABLE XIV
BIOMEDICAL-FEATURE ABLATION (XGBOOST, IDENTICAL LEAKAGE-SAFE SPLITS): PHYSICS FEATURES VS. PHYSICS + BIOMEDICAL ANOMALY FEATURES. DETECTION/PHASE: MACRO-F1; SEVERITY: WITHIN-ONE-LEVEL ACCURACY.

Task (metric)	Protocol	Physics	Physics+Bio	Δ
Detection (macro-F1)	10-fold GK	0.977	0.977	+0.000
Detection (macro-F1)	LOLO	0.948	**0.978**	**+0.030**
Severity (within-1)	10-fold GK	0.997	0.997	+0.000
Severity (within-1)	LOLO	0.942	**0.979**	**+0.037**
Phase (macro-F1)	10-fold GK	0.964	0.964	+0.000
Phase (macro-F1)	LOLO	0.997	0.997	+0.000

0.978 (+0.030) and severity within-one-level accuracy from 0.942 to 0.979 (+0.037), while the already load-robust phase task is unchanged. By gradient-boosting gain the most useful biomedical features are the Hjorth mobility/complexity of the Park modulus, the Higuchi/Petrosian fractal dimension and permutation entropy of the residual, and dispersion entropy—all amplitude-scale-free, which is why they transfer across loads better than absolute-magnitude physics features and directly attack the cross-load severity bottleneck. This cross-domain import of biomedical complexity/irregularity features into stator-current diagnosis is, to our knowledge, new.

I. Feature Importance and Reduction

Importance. Fig. 12 ranks the features that drive the models. For detection, SHAP attribution (a) is led by the Park-vector major axis and eccentricity, the zero/negative-sequence ratios, inter-phase RMS differences, and the EPVA severity factor—all physical inter-turn signatures, confirming the models exploit fault physics rather than dataset shortcuts. Among the novel biomedical features (b), the Park-modulus Hjorth parameters, the residual Higuchi/Petrosian fractal dimension and permutation entropy, and dispersion entropy carry the most weight on the (hardest) severity task, explaining their cross-load gains (Table XIV). *Reduction.* Because many features are redundant or load-dependent, we reduce the bank with the load-invariance-regularized criterion of Section II-F (LIR-mRMR). Fig. 13 compares it with plain mRMR: at a compact k=10 features LIR-mRMR reaches cross-load (LOLO) severity within-one of 0.848 versus mRMR's 0.740 (a) while roughly halving the selected features' across-load drift (b, 0.18 vs 0.38), because it explicitly penalizes features whose class-conditional mean shifts with load. The selected subset is dimensionless and physics-traceable (negative-sequence angle sin/cos, Park eccentricity, $|I_1|$-normalized imbalance, per-phase skew)—a desirable property for safety-critical monitoring—and approaches the full 72-feature accuracy with an order of magnitude fewer inputs. The k=15 dip reflects greedy-selection noise, not a benefit of additional features.

J. Hyperparameter Tuning and Data Augmentation

We finally isolate the effect of leakage-safe hyperparameter tuning and physics-consistent training-fold augmentation on the deployed XGBoost pipeline, evaluated under *10-fold* StratifiedGroupKFold (the 147 operating-point groups satisfy $n_{\text{folds}} \leq n_{\text{groups}}$) and LOLO. Hyperparameters are tuned for cross-load generalization (Section II-F) and frozen for both protocols; augmentation is applied to the training fold only and re-featurised. Table XV reports the three conditions. Tuning alone is marginal on the already-saturated metrics (consistent with the significance analysis of Section IV-D), but *augmentation is the decisive lever for cross-load generalization*: it raises severity within-one-level accuracy from 0.911 to 0.935 under LOLO and from 0.934 to 0.976 in-distribution, cutting

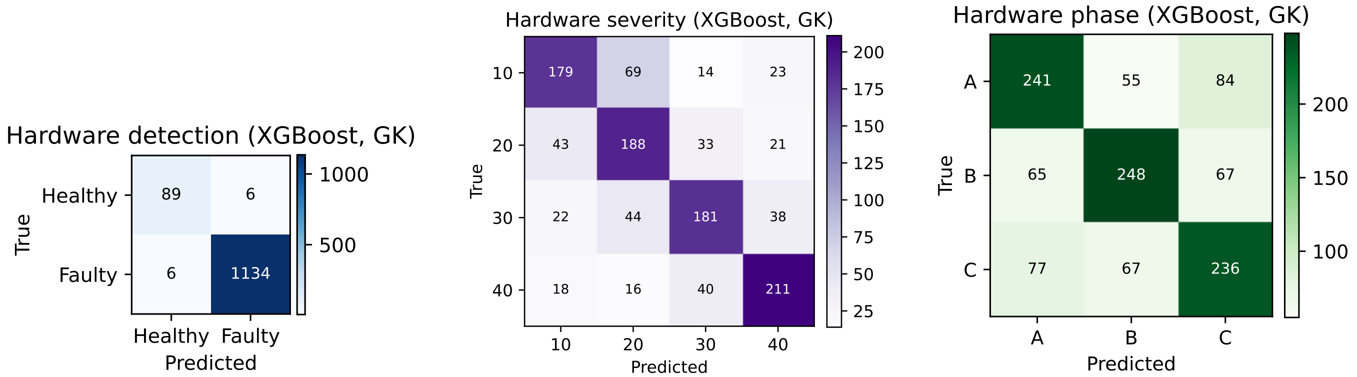

Fig. 11. Within-hardware confusion matrices on the real motor (XGBoost, GroupKFold out-of-fold): detection (left), four-level severity (middle), faulted phase (right). Detection is clean and severity errors are near-diagonal, but the phase matrix shows substantial off-diagonal confusion—the main simulation-to-reality gap.

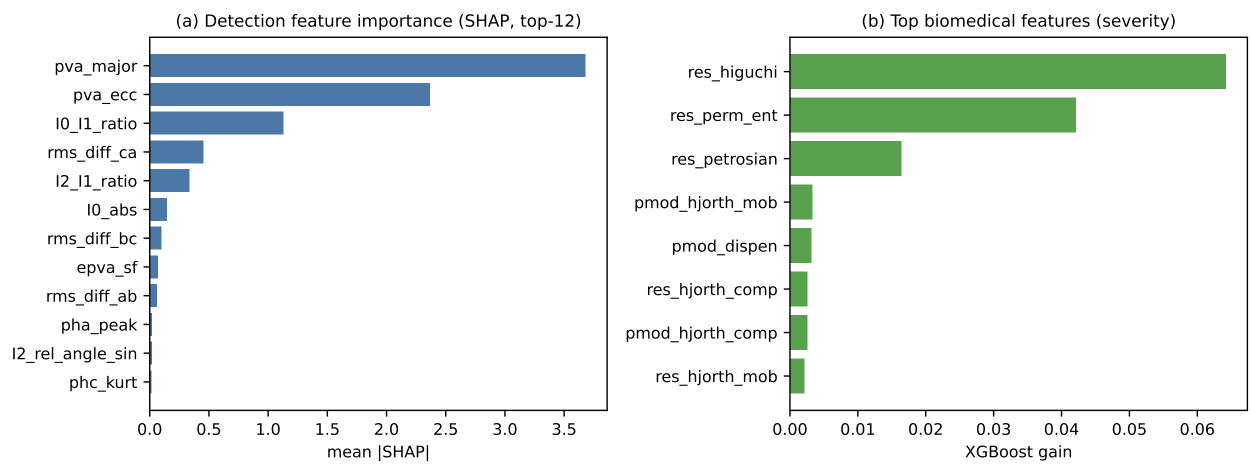

Fig. 12. Feature importance. (a) Stage-1 detection by mean |SHAP| (top-12): Park-vector geometry, sequence-component ratios, inter-phase RMS differences, and the EPVA factor dominate—all physical inter-turn signatures. (b) Most influential *biomedical* features (gradient-boosting gain) on the severity task: Park-modulus Hjorth parameters, residual fractal dimension and permutation entropy, dispersion entropy.

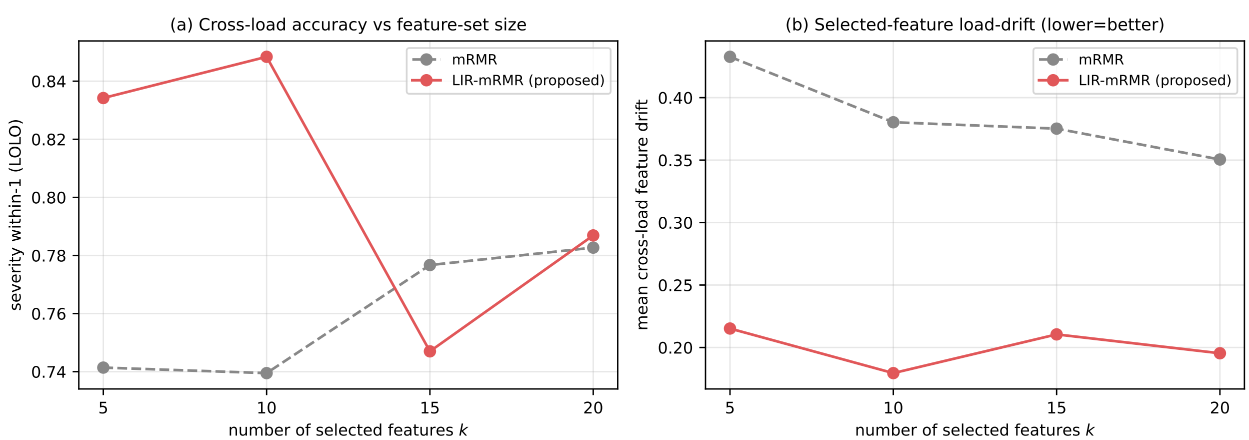

Fig. 13. Load-invariance-regularized feature reduction (LIR-mRMR) vs. plain mRMR, vs. feature-set size k. (a) Cross-load (LOLO) severity within-one accuracy—LIR-mRMR is higher at compact sizes. (b) Mean across-load drift of the selected features (lower is better)—LIR-mRMR roughly halves it by penalising load-unstable features.

the mean absolute error by 20% (0.917 → 0.731 level), and lifts detection LOLO macro-F1 from 0.964 to 0.977, while faulted-phase identification remains at 0.997. This confirms the augmentation principle: synthesising training windows toward unseen operating points—by jointly scaling the three phases, so the inter-phase imbalance and current balance

TABLE XV
Effect of leakage-safe tuning and physics-consistent training-fold augmentation on the deployed XGBoost pipeline (10-fold StratifiedGroupKFold and LOLO). Detection/phase: macro-F1; severity: within-one-level accuracy. Best per row in bold.

Task (metric)	Protocol	Default	Tuned	Tuned+Aug
Detection (macro-F1)	10-fold GK	0.976	0.976	**0.979**
Detection (macro-F1)	LOLO	0.964	0.964	**0.977**
Severity (within-1)	10-fold GK	0.934	0.937	**0.976**
Severity (within-1)	LOLO	0.911	0.908	**0.935**
Phase (macro-F1)	10-fold GK	**0.964**	0.936	**0.964**
Phase (macro-F1)	LOLO	0.994	**0.997**	**0.997**

are preserved—directly improves cross-load diagnosis without fabricating the fault signature.

K. Discussion: Positioning and Deployment

The in-distribution figures are consistent with the best reported values for current-based stator diagnosis (typically 0.96–1.00 accuracy [3], [8], [9]); the contribution here is not a higher in-distribution number—which the saturated benchmark shows is no longer informative—but a demonstration of *which* of those numbers survive a change of operating point, and a set of mechanisms that make the load-dependent ones survive. The in-distribution-versus-LOLO severity gap (within-one 0.99 vs. 0.39 before the load-aware fix) quantifies how much a conventional evaluation would have over-promised; we therefore advocate cross-condition (LOLO) reporting, alongside interpretability evidence, as standard practice. From a deployment standpoint the pipeline is attractive: it consumes only the already-measured three-phase current, runs in well under a millisecond per window on a CPU, exposes interpretable physical quantities (the negative-sequence ratio and EPVA factor that drive a decision), provides calibrated probabilities with an online onset trigger suitable for a protection relay, and offers a foundation-model option (TabPFN-2.5) that is both the most load-robust detector and training-free. Within-hardware validation on a real motor (Section IV-G) confirms that detection transfers (macro-F1 up to 0.99) and severity is usable (within-one $\geq$ 0.86); per-window faulted-phase localization is weak (macro-F1 0.72) but per-recording aggregation—the realistic diagnostic unit—recovers it to 0.93–0.95. The remaining limitations are that the load-robustness results are validated only in simulation (the public real dataset has no load variation), and that a full instrumented campaign with supply-voltage unbalance and richer real severities remains future work.

V. Conclusion

We presented a physics-informed, leakage-safe machine-learning framework for stator inter-turn fault diagnosis in induction motors that performs onset detection, healthy/faulty detection, ordinal severity estimation, and faulted-phase identification from the three-phase stator current. By adopting Leave-One-Load-Out evaluation we showed that detection and faulted-phase identification are intrinsically load-robust, while absolute severity is load-dependent and is recovered to within-one accuracy of 0.95 across unseen loads once the model is made load-aware. A controlled eight-model benchmark—including the tabular foundation model TabPFN-2.5 and an xLSTM sequence network—demonstrated that the in-distribution task is saturated and that cross-load behaviour is the meaningful axis, on which TabPFN-2.5 is the most load-robust detector. Four mechanisms (a sequence-component Stockwell tensor, a load-invariance-regularized feature selection, a FiLM-conditioned multi-task network, and physics-anchored domain adaptation) each independently improved cross-load performance, and the framework was validated simulation-to-experiment on a public real-motor dataset. The end-to-end cascade attains within-one accuracy of 0.95 in-distribution and 0.92 across unseen loads, with calibrated detection, graceful noise degradation, and reliable onset detection. A within-hardware evaluation of all eight models on the real motor confirmed that detection transfers (macro-F1 up to 0.99) and severity is usable; per-window faulted-phase localization is the main gap (macro-F1 0.72) but per-recording aggregation recovers it to 0.93–0.95. Future work includes closing the residual per-window phase gap (higher sampling, supervised phase calibration), a stronger conditional generator for physics-constrained augmentation (the signature-preserving training-fold augmentation introduced here already cut cross-load severity error by 20%), full selective-state-space and Kolmogorov–Arnold backbones, an experimental campaign with supply voltage-unbalance rejection, and group-conditional conformal calibration of the severity intervals. All code, dataset splits, and result artifacts are released for reproducibility.

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